MANJUSHA GUNTUR

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TECHNICAL LEAD | SOFTWARE ARCHITECT

SOFTWARE ARCHITECTURE & DESIGN, FULL-STACK DEVELOPMENT, CLOUD COMPUTING

Expertise in designing and developing secure software solutions.

Technical Lead and Forward-Deployed Engineer with 15+ years of experience designing, building, and delivering enterprise applications across financial services, government, retail, and technology domains. I have strong hands-on experience in Java, Spring Boot, micro-services, cloud platforms, Kafka, database optimization, and full-stack application development. In my recent work, I have partnered closely with business, product, architecture, and engineering teams to convert real-world requirements into production-ready systems. I have led delivery from requirements and design through build, testing, release, and production support, with a focus on reliability, scalability, and clean engineering practices. Recently, I have also used AI agents for pull request code reviews, SQL script generation/review, technical analysis, and developer productivity, along with MCP Server for Confluence updates and Jira story creation.

Areas of Expertise

Object Oriented Design & Analysis • Microservices & Distributed Systems • Agile & Scrum Methodologies
Multi-Tier Enterprise Applications Creation • Software Development Life Cycle • Use Case Diagrams Creation
Database Management • Service-Oriented Architecture • Pair Programming • Technical Solutions Designing
Project Planning & Execution • Application Integrations & Migrations • Test-driven Development

Technical Skills

Java• Scala • Groovy • Python • PL/SQL • UNIX • Kafka • JavaScript • Terraform • Angular • NodeJS • Spring • Maven • Gradle • Easy Mock • Mockito • JUnit • Elastic Search • Log Stash • Kibana • Redis • Hibernate • JPA • Oracle • SQL Server • DB2 • Postgres • MongoDB • MySQL • Jboss • BEA WebLogic • WebSphere • Tomcat • Jenkins • CSS HTML • GIT • Bitbucket • SVN • Perforce AccuRev • Jira • Confluence • GCP • PCF • AWS • Azure • SonarQube • Docker • Solr • Splunk • MCP Server •AI-Native Software Engineering

EDUCATION

Post Graduate Program in AI and Machine Learning, McCombs School of Business, *Expected November 2026*

Master of Science (MS) in Computer Science, Northern Illinois University, Dekalb, IL

Bachelor of Technology (B. Tech) in E&E Engineering, Jawaharlal Nehru Technological University, India

CERTIFICATIONS

Pivotal Certified Spring Professional | Project Management Professional | Scrum Master | PMI Agile Certified Practitioner | Certified Scrum Product Owner | Student Pilot at Randon Aviation

TECHNICAL PUBLICATIONS

Building an AI Native Software Engineering Platform | Hashnode & Hackernoon, July 2026

Can Agentic AI Catch Architecture Flaws Before Implementation? | Hashnode & Hackernoon, July 2026

I Built an AI Code Review Agent That Scales Architect-Level Standards Across Every PR| Hackernoon, August 2026

Why Idempotency Matters More Than Retries in Microservices| Hackernoon, August 2026

Scaling a Kafka Consumer From 4K to 25K Events per Sec While Preserving Ordering| Hackernoon, August 2026

SELECTED ARCHITECTURE AND AI PROJECTS

SmartCodeReview | AI Code Review Agent AUG 2026
Built an AI code review agent around a Java Spring Boot Order Management microservice to apply developer and architect standards before human review. It evaluates code quality, architecture, security, reliability, testing, and production readiness while separating implementation defects from architect owned decisions.

The Cosmic Pattern | Multi Agent Product and AI Architecture AUG 2026 - Present
Independently designed and launched a Vedic astrology publishing platform with an architecture roadmap for community capabilities and a personalized AI assistant combining chart calculations, curated knowledge, RAG, and conversationa AI. The product roadmap explores a multi agent software engineering model across requirements, architecture, development, testing, security, and release as the platform is incrementally implemented.

Distributed Systems Design | Architecture Portfolio AUG 2026 - Present
Building ten production scale distributed system designs, each presented as both a detailed interview study and an architect portfolio case study. Every design covers requirements, capacity, APIs, data models, architecture, critical flows, scaling, reliability, failure recovery, multi region design, and engineering tradeoffs.

PROFESSIONAL EXPERIENCE

CHARLES SCHWAB – Austin, TX JUN 2025 – Present
Technical Lead/ Technical Project Manager – Enterprise Segmentation and Treatments
Led enterprise segmentation and treatment eligibility modernization by building Kafka-based consumer applications and micro-services with production operational readiness for near real-time eligibility processing.

- Delivered 4 **production applications from scratch in 12 months**, owning end-to-end execution from requirements, design, build, testing, release, deployment coordination, and production readiness.
- Operated in a Forward-Deployed Engineering style, partnering with product, architecture, business, QA, and engineering teams to translate eligibility requirements into scalable production systems.
- Used AI agents for pull request code reviews, SQL script generation/review, technical analysis, and developer productivity workflows, and used MCP Server to update Confluence documentation and create jira stories.
- Led and coordinated an 11-member cross-functional onsite/offshore team through sprint planning, prioritization, dependency management, implementation, and release readiness.
- Contributed hands-on using Java, Spring Boot, Kafka, PostgreSQL, REST APIs, Gradle, Bamboo CI/CD, and Splunk.
- Designed Kafka consumer patterns, data-flow decisions, retry/backoff handling, idempotency/deduplication, structured exception handling, and downstream integration patterns.
- Established performance/SLO targets for Kafka consumers: **p95 throughput ~15,000 events/sec, 100% consumption success rate**, and **CPU/Memory ≤70% at p95**.
- Leveraged and managed cloud resources, ensuring consistency and repeatability across environments.
- Built Splunk dashboards and queries for throughput, processing failures, consumer lag, CPU/memory usage, runtime validation, triage, and production support.
- Designed resiliency patterns (retry/backoff, idempotency/dedup) to protect downstream systems and improve processing correctness under spikes.
- Partnered with product/architecture/business to define eligibility rules and author/review PRS, HLD/LLD, and design decisions aligned with enterprise standards.
- Optimized PostgreSQL for classification workloads (indexes, query tuning, views/materialized views) and improved release confidence with JUnit/Mockito/Cucumber + Bamboo/Gradle CI/CD.

STATE OF UTAH – Salt Lake City, UT 2022 – JUN 2025
Technical Architect/Developer – Utah Works
Gather and analyze functional business requirements in alliance with the product owner, designing and implementing effective solutions for streamlined development. Lead the development phase using Spring Framework and Angular, including a full upgrade of applications to Angular 17. Apply design patterns (Singleton, Façade, Builder, Spring MVC, DAO) to improve application architecture and maintainability. Implement and deploy micro-services-based applications on AWS.

- Expertly engineered and optimized stored procedures, significantly improving data processing efficiency for eREP and Workshop Interfaces.
- Built and maintained RESTful web services utilizing Spring REST, Batch, Security, Data JPA, and Boot.
- Strategically led proof-of-concept development and implementation of complex micro-services architecture, particularly for the high-impact eREP module, a financial support system for employment seekers.
- Leveraged infrastructure-as-code (IaC) tools such as Terraform to provision and manage cloud resources, ensuring consistency and repeatability across environments.

ARROW ELECTRONICS, INC. – Denver, CO 2019 – 2021

Applications Engineer II – Sales Workspace Global

Improved overall application architecture and streamlined the development process.

- Engineered a seamless Oracle ADF migration for Sales Workspace Global, transforming the application into a centralized quoting & ordering system and enabling the global manufacturers, B2B clients, and end users to access quotes & pricing efficiently using Spring modules, JPA and Hibernate.
- Led an offshore team and migrated applications in Arrow ECS AngularJS and other Angular versions projects to Angular 11.
- Developed POC's as part of Architecture team in Arrow ECS on GraphQL and Redis cache.
- Integrated Redis Cache and Azure role-based access control for Sales Workspace Global.
- Integrated monitoring and logging solutions using tools like Prometheus, Grafana, and ELK Stack, enabling real-time performance insights and proactive incident management.

STATE OF UTAH – Salt Lake City, UT 2018 – 2019, 2021

Technical Architect/Lead Developer – Utah Data Research Center

Collaborated with product owners to document business requirements, producing comprehensive functional, technical, migration, and execution specifications in Confluence. Wrote complex stored procedures for a matching algorithm to integrate data from multiple systems; implemented data de-identification processes for researcher access. Created and mapped domain objects to Oracle 12c tables using JPA, implemented design patterns to enhance maintainability. Built Batch job modules to process and load partner data from zip files into raw tables with custom transformations.

- Converted Oracle PL/SQL procedures to Python programs and migrated schemas, tables, and procedures from Oracle to Postgres for enhanced flexibility in cloud environments.
- Strategically led the migration of critical applications to GCP, enhancing cloud infrastructure and operational efficiency.
- Developed secure modules from scratch & crafted PL/SQL procedures in Oracle aimed at ensuring data anonymization.
- Led the successful migration of the UDRC backend application from on-premise to GCP independently, completing a 10-month, 4-developer project in just 6 months.
- Played dual roles as Senior Developer and Project Architect, designing and overseeing the migration while navigating new GCP tools and architecture with rapid learning and execution.
- Expertly built and orchestrated containerized applications with Docker and Kubernetes, achieving high availability and scalability.

NELNET, INC. – Madison, WI 2018

Sr. Java/Java EE Lead Developer – PMC Build Out Full-Service Solution

Analyzed, developed, and implemented complex functionalities in alignment with business requirements.

- Developed front-end functionalities with AngularJS and integrated ElasticSearch for enhanced search capabilities.
- Integrated SonarQube for our application for continuous code quality inspections.

CHARLES SCHWAB & CO, INC. – Phoenix, AZ 2016 – 2017

Sr. Java/Java EE Lead Developer – BDE /Bank Data Interface (BDI)

Introduced and implemented micro-services architecture to the Bank Data Exchange, streamlining operations and enabling efficient data handling. Led GitHub migration and trained a team of 11, enhancing version control practices across the team.

MACY'S SYSTEMS AND TECHNOLOGY – Lorain, OH 2014 – 2016

Sr. Java/Java EE Lead Developer/Scrum Master – OmniClient

Led the migration of the Omniclient register software, implementing Spring Batch to migrate data from GoldenGate and other sources to Oracle DB. Introduced and implemented the Spring MVC in the OmniClient Application, which was a modernization project migrated from traditional java application.

JOHN DEERE – Des Moines, IA 2013 – 2014
Sr. Java/Java EE Lead Developer
Headed the MyJohnDeere Platform Services Software solution that provided services for John Deere Portal web applications, and Maintenance App portal that monitored machine status, mobility, and condition of the machines for individual users and dealers. Utilized Spring Framework and Restful web services for the project implementation.

NETAPP – Waltham, MA 2013
Sustaining Engineering Lead
Led the OnCommand Balance project, which provided performance planning, troubleshooting, and optimization for physical and virtual data centers to help manage the performance and workloads of dynamic shared infrastructure. Provided custom support for clients and incorporated new features asked for their in their custom software solutions.

VERIZON BUSINESS – Tampa, FL 2010-2013
Java Developer
Developed Provisioning controller which is a workflow system implemented to migrate workflow, provisioning from the Mainframe system FNE. The main goal of his project is to shut down this FNE and automate the task assignment. With this PC, the business users are mainly encouraged to use PC and UPI (user Interface)thus enabling the users to migrate from FNE to PC.